

PROGRAM 1: PHP Program to Remove Duplicates from a Sorted List

Program 1 – Viva Questions & Answers (10)

1. What function in PHP removes duplicate values from an array?

A: `array_unique()` is used to remove duplicates.

2. Does `array_unique()` work only on sorted arrays?

A: No, it works on any array, but sorted arrays give predictable results.

3. What data type does `array_unique()` return?

A: It returns an array.

4. Why is a sorted array preferred for removing duplicates?

A: Because duplicates appear next to each other, making detection easier.

5. What happens to the keys after using `array_unique()`?

A: Keys are preserved unless you use `array_values()`.

6. Can we remove duplicates manually without using a function?

A: Yes, using loops and checking previously visited values.

7. What is the time complexity of removing duplicates from a sorted list manually?

A: $O(n)$

8. What is an indexed array?

A: An array with numeric keys starting from 0.

9. What is the use of `print_r()` in PHP?

A: It prints arrays in human-readable format.

10. Why is removing duplicates important?

A: To avoid redundant data and improve performance.

PROGRAM 2: PHP Script to Print Pattern

Program 2 – Viva Questions & Answers (10)

1. Why do we use nested loops to print patterns?

A: Outer loop controls rows, inner loop controls columns.

2. What is the purpose of the `<br>` tag in pattern printing?

A: To move to the next line in HTML output.

3. Can we print patterns without loops in PHP?

A: Yes, but it is inefficient and not recommended.

4. What does `echo "*" "` do in a loop?

A: Prints an asterisk followed by a space.

5. What loop types can be used in PHP for pattern printing?

A: `for`, `while`, and `do...while`.

6. Why is PHP used to print patterns in web development?

A: For learning logic; rarely used in real projects.

7. What is the difference between `echo` and `print`?

A: `echo` is slightly faster, and can output multiple parameters.

8. How do we control the number of rows in a pattern?

A: By adjusting the condition of the outer loop.

9. What is a decreasing pattern?

A: A pattern where row length decreases with each level.

10. Why is pattern printing useful?

A: Helps students understand loops and nested loops.

PROGRAM 3: PHP Function to Generate Captcha Code

Program 3 – Viva Questions & Answers (10)

1. What is a CAPTCHA?

A: CAPTCHA is a test to ensure the user is human and not a bot.

2. Why do we use random characters in a CAPTCHA?

A: To prevent automated bots from guessing the pattern.

3. What function is used to shuffle characters in PHP?

A: `str_shuffle()`.

4. How do we extract a fixed-length string in PHP?

A: Using `substr()`.

5. What is the typical length of a CAPTCHA code?

A: Usually 4–6 characters.

6. Why do CAPTCHAs use both letters and numbers?

A: To increase complexity and security.

7. Can CAPTCHA be generated without PHP?

A: Yes, with JavaScript, Python, or server libraries.

8. What is the purpose of a CAPTCHA?

A: To block spam, bots, and automated form submissions.

9. Are CAPTCHAs case-sensitive?

A: They can be, depending on implementation.

10. Why is server-side CAPTCHA generation important?

A: Because client-side CAPTCHA can be manipulated or bypassed

CAPTCHA-Completely Automated Public Turing test to tell Computers and Humans Apart

PROGRAM 4: PHP Program to Display Today's Date

Program 4 – Viva Questions & Answers (10)

1. Which function in PHP displays current date?

A: `date()` function.

2. What format does `date("d-m-Y")` follow?

A: Day-Month-Year.

3. What is the role of `y` in the date format?

A: It prints the year in 4 digits.

4. Can PHP display time also?

A: Yes, using `date("H:i:s")`.

5. Which function returns the current timestamp?

A: `time()`.

6. What does `date_default_timezone_set()` do?

A: Sets the timezone for date/time functions.

7. Is the date function server-dependent?

A: Yes, it uses the server's clock.

8. How do you show the full weekday name?

A: `date("l")` (lowercase L).

9. What is the output of `date("d/m/Y")`?

A: Date in `dd/mm/yyyy` format.

10. Can PHP format dates from database values?

A: Yes, using `strtotime()` to convert and then `date()`.

PROGRAM 5: Factorial of a Number

Viva Questions & Answers (10)

1. What is a factorial?

A: Product of all natural numbers from 1 to n.

2. Which loop is commonly used to find factorial?

A: `for` loop.

3. Is factorial defined for negative numbers?

A: No, factorial is only for non-negative integers.

4. What is the factorial of 0?

A: 1 (by definition).

5. Can recursion be used for factorial?

A: Yes, very commonly used.

6. Why does factorial grow very fast?

A: Because it is multiplicative.

7. What data type is required for large factorial?

A: float or big integer libraries.

8. Is factorial used in mathematics?

A: Yes, in probability & permutations.

9. What operator is used for multiplication?

A: `*`

10. Where is factorial used in programming?

A: Combinations, permutations, recursion practice.

PROGRAM 6: Change Background Color Based on Day of Week

Viva Questions & Answers (10)

1. Which PHP function retrieves the current day?

A: `date("l")`.

2. What data structure is used for storing colors?

A: An associative array.

3. How do we insert PHP inside HTML?

A: Using `<?= ?>` or `<?php ?>`.

4. Why use if–else or array for this program?

A: To match days with colors.

5. Can CSS be dynamically controlled in PHP?

A: Yes, using inline styles.

6. What is an associative array?

A: An array with named keys.

7. How many days exist for mapping?

A: 7 days.

8. Why is background color changed?

A: For UI enhancement and learning.

9. What is the use of `style=""` attribute?

A: To apply inline CSS.

10. Is this program server-side or client-side?

A: Server-side (PHP).

PROGRAM 7: Generate Prime Numbers

Viva Questions & Answers (10)

1. **What is a prime number?**

A: A number divisible only by 1 and itself.

2. **Is 1 a prime number?**

A: No.

3. **Why check up to `sqrt(n)` for primality?**

A: To reduce time complexity.

4. **Which loop checks divisibility?**

A: Inner `for` loop.

5. **What happens if a number is divisible by any `i`?**

A: It is not prime.

6. **What is the smallest prime number?**

A: 2.

7. **Why do we use boolean flags?**

A: To mark whether a number is prime.

8. **What does `%` operator do?**

A: Finds remainder.

9. **Can arrays store prime numbers?**

A: Yes.

10. **Where are primes used?**

A: Cryptography and algorithms.

PROGRAM 8: Area of a Circle

Viva Questions & Answers (10)

1. **What is the formula for area of a circle?**

A: πr^2

2. **Which constant is used for π ?**

A: 3.1416 or `pi()` function.

3. **What is the data type of radius?**

A: Integer or float.

4. **Can PHP calculate with decimals?**

A: Yes.

5. **What operator is used for exponent?**

A: `*` ($r \times r$)

6. **Is math library required?**

A: No, simple operations are enough.

7. **Does negative radius make sense?**

A: No.

8. **What is the output type?**

A: Float value.

9. **Can input be taken from form?**

A: Yes.

10. **Which function formats decimal places?**

A: `number_format()`.

PROGRAM 9: Input Validation in PHP

Viva Questions & Answers (10)

1. Why is input validation necessary?

A: To prevent errors and security issues.

2. What does `$_POST` contain?

A: Form data sent via POST method.

3. What is `trim()` used for?

A: Removing extra spaces.

4. What is `htmlspecialchars()` used for?

A: Preventing HTML injection.

5. What type of error occurs if input is empty?

A: Validation error.

6. Why use server-side validation?

A: More secure than client-side.

7. What method attribute is used in forms?

A: POST.

8. Which tag submits the form?

A: `<input type="submit">`

9. What function checks if a variable is empty?

A: `empty()`.

10. What is form sanitization?

A: Cleaning user inputs.

PROGRAM 10: Evaluate Mathematical Expression Using JavaScript

Viva Questions & Answers (10)

1. What JS function evaluates expressions?

A: `eval()`.

2. How do we read input values?

A: `document.getElementById().value`

3. Where does JS execute?

A: In the browser (client-side).

4. Can eval handle decimal operations?

A: Yes.

5. What does innerHTML do?

A: Inserts content into an element.

6. What event triggers calculation?

A: Button click.

7. Can JavaScript validate expressions?

A: Partially; needs custom checks.

8. What type of value is entered in the text box?

A: String.

9. Who converts the string to expression?

A: The `eval` interpreter

PROGRAM 11: Page with Dynamic Effects (Layers + Animation)

Viva Questions & Answers (10)

1. What is CSS animation used for?

A: To create movement and visual effects on web pages.

2. Which CSS rule defines animations?

A: `@keyframes`.

3. What does the `animation` property control?

A: Duration, timing, and behavior of the animation.

4. What is a layer in web design?

A: An element with relative or absolute positioning.

5. What property positions elements freely?

A: `position: absolute` OR `position: relative`.

6. What is the use of the `left` property in animation?

A: Moves an element horizontally.

7. Can animations loop?

A: Yes, using `infinite` keyword.

8. Is JavaScript required for CSS animations?

A: No, CSS alone can animate objects.

9. Why do we use `<div>` elements?

A: To create containers for animation.

10. What is the difference between transition and animation?

A: Transition occurs on event; animation runs automatically.

PROGRAM 12: JavaScript Program to Find Sum of N Natural Numbers

1. What is a natural number?

A: Positive integers starting from 1.

2. Which loop is used for summation?

A: Typically a `for` loop.

3. What variable stores the sum?

A: A counter variable like `sum`.

4. What is the formula for sum of N natural numbers?

A: $n(n+1)/2$

5. What data type are loop counters in JS?

A: Number.

6. What method outputs text in JS?

A: `document.write()` or `innerHTML`.

7. What is the initial value of sum?

A: 0.

8. Can the sum be calculated using a function?

A: Yes.

9. Which operator is used for addition?

A: +

10. Can input be taken from the user?

A: Yes, using `prompt()` or forms.

PROGRAM 13: PHP Exception Handling – Date Format Checking

Viva Questions & Answers (10)

1. What is exception handling?

A: Managing runtime errors using try-catch blocks.

2. Which keyword throws an exception?

A: `throw`.

3. Which block catches errors?

A: `catch`.

4. What function checks valid date format?

A: `strtotime()`.

5. What class is used for exceptions in PHP?

A: `Exception`.

6. What happens if a date is invalid?

A: Exception is thrown and caught.

7. What is the purpose of `try{} block`?

A: To test potentially risky code.

8. Why validate dates?

A: To avoid incorrect calculations & errors.

9. Which function displays an error message?

A: `$e->getMessage()`.

10. Is exception handling mandatory?

A: No, but highly recommended.

PROGRAM 14: Image Switch Using Mouseover & Mouseout

1. Which event occurs when the mouse moves over an element?

A: `onmouseover`.

2. Which event triggers when the mouse leaves?

A: `onmouseout`.

3. What object is affected by these events?

A: HTML `<img>` tag.

4. Which attribute changes the image?

A: `src`.

5. Can multiple images be switched?

A: Yes.

6. Where do JavaScript events run?

A: On the client-side.

7. Are event handlers case-sensitive?

A: Yes.

8. Why is this effect used?

A: For interactive UI elements.

9. Can CSS alone switch images?

A: Yes, using hover pseudo-class.

10. Is JavaScript required in inline attributes?

A: No external JS needed.

PROGRAM 15: Student Form – JS Total, Average, Grade

Viva Questions & Answers (10)

1. What input type is used for marks?

A: `text` or `number`.

2. How do we convert string to integer?

A: `parseInt()`.

3. How is total calculated?

A: Sum of all subject marks.

4. What is the formula for average?

A: Total / number of subjects.

5. What is grading based on?

A: Average marks.

6. What function displays output?

A: `innerHTML`.

7. What is the role of a button?

A: Trigger calculation.

8. Can negative marks exist?

A: No, validation needed.

9. Where does JavaScript run?

A: Browser.

10. Why calculate grade on client side?

A: Fast and interactive.

PROGRAM 16: Employee Form – DA, HRA, PF, TAX, Net Pay Calculation

Viva Questions & Answers (10)

1. What is DA?

A: Dearness Allowance — a percentage of basic salary.

2. What is HRA?

A: House Rent Allowance.

3. What is PF?

A: Provident Fund deduction.

4. What is Gross Pay?

A: Basic + Allowances.

5. What is Net Pay?

A: Gross Pay – Deductions.

6. Which JavaScript function converts text to float?

A: `parseFloat()`.

7. Why use JavaScript for salary calculation?

A: To instantly show results.

8. What are deductions?

A: PF + TAX.

9. Which operator is used for percentage?

A: `*` and decimals (e.g., 0.10).

10. Where is salary data finally displayed?

A: Inside an HTML element.

PROGRAM 17: Fibonacci Series in PHP

Viva Questions & Answers (10)

1. What is Fibonacci sequence?

A: Series where each number = sum of previous two numbers.

2. What are the first two Fibonacci numbers?

A: 0 and 1.

3. Which operator performs addition?

A: +.

4. Which loop is commonly used?

A: `for`.

5. Where is Fibonacci used?

A: Algorithms, math, nature patterns.

6. What does the variable `c` store?

A: New Fibonacci number.

7. Is recursion possible for Fibonacci?

A: Yes, but slower.

8. Why store `a` and `b` separately?

A: To update previous values.

9. What is the stopping condition?

A: Loop limit.

10. Is Fibonacci infinite?

A: Yes, but we print a limited range.

PROGRAM 18: PHP Searching of Data by Criteria

Viva Questions & Answers (10)

1. Which function checks if value exists in array?

A: `in_array()`.

2. What data type stores list of values?

A: Array.

3. Is search case-sensitive?

A: Yes.

4. What does search return?

A: Found / Not Found.

5. Can searching be done in unsorted arrays?

A: Yes.

6. What is linear search?

A: Checking elements one by one.

7. Can we search using loops?

A: Yes.

8. What does `echo` output?

A: Result message.

9. Where is searching used?

A: Databases, file systems, applications.

10. Is PHP search fast for large datasets?

A: Not always; better to use database queries.

PROGRAM 19: PHP Cookies – Set & Retrieve

Viva Questions & Answers (10)

1. What is a cookie?

A: Small data stored on the user's browser.

2. Which function creates a cookie?

A: `setcookie()`.

3. What superglobal retrieves cookies?

A: `$_COOKIE`.

4. What is cookie expiry time?

A: Duration after which cookie is deleted.

5. Where are cookies stored?

A: Client-side browser.

6. Can cookies store multiple values?

A: Yes, as arrays or JSON.

7. Are cookies secure?

A: Not fully; can be modified.

8. Are cookies sent with each request?

A: Yes.

9. Can cookies be disabled?

A: Yes by user.

10. Difference between session and cookie?

A: Session is server-side; cookie is client-side.

PROGRAM 20: Simple PHP Website for a College

Viva Questions & Answers (10)

1. What is a website?

A: A collection of web pages.

2. Which tag displays headings?

A: `<h1>`.

3. Is PHP used for structure or logic?

A: Logic and dynamic content.

4. What is HTML?

A: Markup language to design web pages.

5. Can PHP and HTML be mixed?

A: Yes.

6. Why use echo for page content?

A: To output dynamic text.

7. What is a home page?

A: First page of a website.

8. What file extension do PHP pages use?

A: `.php`.

9. Can PHP fetch college data from database?

A: Yes.

10. What is the purpose of this program?

A: To demonstrate dynamic page creation.

PROGRAM 21: PHP Read & Write File Using Form

Viva Questions & Answers (10)

1. Which PHP function writes data to a file?

A: `file_put_contents()`.

2. Which function reads file content?

A: `file_get_contents()`.

3. Why use textarea?

A: To allow large text input.

4. What method submits the form?

A: POST.

5. What is a file?

A: Data stored on server.

6. Where are files stored?

A: In project directory.

7. Is file handling server-side or client-side?

A: Server-side (PHP).

8. Why use `nl2br()` ?

A: Converts newlines to `<br>`.

9. Can PHP create new files?

A: Yes.it can create using file function

10.Is file handling safe?

A: Only if permissions are properly set.